

A TIGHT FAMILY FOR THE STATE COMPLEXITY OF LINEAR SUBSEQUENCES OF AUTOMATIC SEQUENCES

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ABSTRACT. Open Problem 1 of Moradi, Rampersad and Shallit (arXiv:2512.10017v5, §4) asks for a class of automatic sequences $(h_m(i))_{i \geq 0}$, each generated by an m -state msd-first DFAO, whose linear subsequences $(h_m(ni))_{i \geq 0}$ have state complexity at least Cnm^2 for infinitely many n and some constant C . We answer it with the generalised Thue–Morse family $h_m(i) = s_2(i) \bmod m$ (s_2 the binary digit sum): for every $m \geq 2$ and odd $n \geq 3$ the minimal msd-first DFAO of $(h_m(ni))_{i \geq 0}$ has exactly $m^2(n-1) - m \max(2^{p-1}, r)$ states, where $n-1 = 2^p + r$, $0 < r \leq 2^p$. This is at least $\frac{1}{2}nm^2$ for every odd $n \geq 3$, and $\limsup_n \text{sc}(h_m(ni))/(nm^2)$ equals $1 - 1/(3m) \geq 5/6$; the value $C = 1/2$ is attained exactly at $(m, n) = (2, 3)$, so within this family it is not improvable uniformly over odd n . The proof joins three ingredients: a Myhill–Nerode lemma for every digit-additive sequence and every n , generalising the authors’ own optimality theorem; a reachability lemma for odd n from Gelfond’s 1968 equidistribution theorem, replacing their Thue–Morse-specific one; and the known factor complexity of the generalised Thue–Morse word.

1. BACKGROUND AND THE PROBLEM

Fix a base $k \geq 2$ and write $\Sigma_k = \{0, \dots, k-1\}$; for $x \in \Sigma_k^*$ let $[x]_k$ be the integer it represents, most-significant digit first, and for $i \in \mathbb{N}$ let $(i)_k$ be its leading-zero-free representation. A sequence $h = (h(i))_{i \geq 0}$ over a finite alphabet Δ is k -automatic if $h(i) = \tau(\delta(q_0, (i)_k))$ for some DFAO $M = (Q, \Sigma_k, \delta, q_0, \Delta, \tau)$. Input is msd-first with leading zeros allowed throughout, so we assume M is *zero-consistent*: $\delta(q_0, 0) = q_0$, which is exactly what makes the recursion $h'(ki+b) = \delta(h'(i), b)$ valid at $i = 0$ as well. The *interior sequence* is $h'(i) = \delta(q_0, (i)_k) \in Q$, generated by the *intrinsic* DFAO $\widehat{M} = (Q, \Sigma_k, \delta, q_0, Q, \text{id})$. We write $\rho_x(n)$ for the number of length- n factors of x and $\text{sc}(x)$ for the number of states of the minimal msd-first DFAO generating x .

Moradi, Rampersad and Shallit [6] relate these two quantities. For $n \geq 1$ and $0 \leq c < n$, their Theorem 9(a) generates $(h(ni+c))_{i \geq 0}$ by a DFAO whose states are length- n factors of h' , the state reached on input x being the window $[h'(n[x]_k), \dots, h'(n[x]_k + n-1)]$ with output $[r_0, \dots, r_{n-1}] \mapsto \tau(r_c)$. Hence $\text{sc}(h(ni+c)) \leq \rho_{h'}(n)$; with $\rho_x(n) \leq kn \rho_{\widehat{M}}(2) \leq knm^2$ [1, Thm. 10.3.1] this gives their Corollary 10(a): $\text{sc}(h(ni+c)) \leq knm^2$. They then write [6, §4]:

We do not know if the bounds in Corollary 10 are tight. This seems worthy of further study, so we state it as an open problem.

Open Problem 1 ([6, §4]). Find a class of automatic sequences $(h_m(i))_{i \geq 0}$, generated by a DFAO of m states with msd-first input, such that there is a constant C so the linear subsequences $(h_m(ni))_{i \geq 0}$ have at least the state complexity Cnm^2 for infinitely many n .

(Quoted verbatim from arXiv:2512.10017v5, 2 April 2026. It appears again, verbatim, as Problem 1 of Moradi’s MMath thesis [5], whose “More Open Problems” section leaves it unresolved, so the problem is open as of its author’s latest treatment.)

For $m = 2$ the authors settle the state count: their Theorem 16 shows the Theorem 9 construction optimal for the Thue–Morse word t and odd n ; their Lemma 15, quoting [2], shows every length- n factor of t occurs at a position divisible by odd n ; and with Brlek’s factor complexity of t [3] this yields $\text{sc}(t(ni)) = \rho_t(n/\nu_2(n))$ (their Theorem 14) and the explicit Corollary 17. What is missing is the m -dependence — a family in which the m^2 of Corollary 10 is visible.

2. THE FAMILY

For $m \geq 2$ let $s_2(i)$ be the number of 1s in the binary representation of i , and let

$$t_{2,m} = (s_2(i) \bmod m)_{i \geq 0}$$

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be the *generalised Thue–Morse word* over $\mathbb{Z}/m\mathbb{Z}$; for $m = 2$ this is the Thue–Morse word. It is generated by the m -state zero-consistent msd-first DFAO

$$M_m = (\mathbb{Z}/m\mathbb{Z}, \{0, 1\}, \delta, 0, \mathbb{Z}/m\mathbb{Z}, \text{id}), \quad \delta(q, d) = q + d \bmod m,$$

equivalently by the 2-uniform morphism $j \mapsto j(j+1)$ on $\mathbb{Z}/m\mathbb{Z}$ with identity coding. State q is reached by 1^q and the coding is injective, so distinct states are separated by the empty word: M_m is minimal, $\text{sc}(t_{2,m}) = m$, and the interior sequence of M_m is $t_{2,m}$ itself. So $t_{2,m}$ is a legitimate candidate for Problem 1. The one structural property we use is that it is *digit-additive*.

Definition 2.1. A sequence $h : \mathbb{N} \rightarrow \mathbb{Z}/m\mathbb{Z}$ is *k-digit-additive* if

$$h(k^L X + y) \equiv h(X) + h(y) \pmod{m} \quad \text{for all } X \geq 0, L \geq 0, 0 \leq y < k^L. \quad (1)$$

Equivalently, $h(i) = \sum_j w_{d_j} \bmod m$ is a weighted digit sum with weights $w : \Sigma_k \rightarrow \mathbb{Z}/m\mathbb{Z}$ satisfying $w_0 = 0$: the representation of $k^L X + y$ is that of X followed by the zero-padded length- L representation of y , and $w_0 = 0$ makes the padding harmless. Condition (1) forces $h(0) = 0$, and the associated DFAO $\delta(q, d) = q + w_d$ is zero-consistent. Here $t_{2,m}$ is the case $k = 2$, $(w_0, w_1) = (0, 1)$.

3. LEMMA A: NERODE CLASSES ARE EXACTLY THE WINDOWS

Lemma 3.1 (Lemma A). *Let $k \geq 2$, $m \geq 2$, let $h : \mathbb{N} \rightarrow \mathbb{Z}/m\mathbb{Z}$ be k -digit-additive, and let $n \geq 1$. For $j \geq 0$ put*

$$W(j) = (h(nj + g))_{0 \leq g < n} \in (\mathbb{Z}/m\mathbb{Z})^n.$$

Then two words $x, y \in \Sigma_k^$ are Myhill–Nerode equivalent for the sequence $(h(ni))_{i \geq 0}$ if and only if $W([x]_k) = W([y]_k)$. Consequently*

$$\text{sc}(h(ni)) = \#\{W(j) : j \geq 0\},$$

that is, the reachable part of the Theorem 9(a) automaton of [6] is already minimal.

Proof. Recall that $x \sim y$ means $h(n[xz]_k) = h(n[yz]_k)$ for every $z \in \Sigma_k^*$.

Sufficiency. Let $z \in \Sigma_k^L$, so $n[xz]_k = k^L \cdot n[x]_k + n[z]_k$. Write $n[z]_k = k^L g + w$ with $0 \leq w < k^L$ and $g = \lfloor n[z]_k / k^L \rfloor$; since $[z]_k \leq k^L - 1$ we have $n[z]_k < nk^L$, hence $g \leq n - 1$. Thus $n[xz]_k = k^L(n[x]_k + g) + w$, and (1) gives

$$h(n[xz]_k) = h(n[x]_k + g) + h(w) = W([x]_k)[g] + h(w) \pmod{m}. \quad (2)$$

The right-hand side depends on x only through $W([x]_k)$, so $W([x]_k) = W([y]_k)$ implies $x \sim y$.

Necessity. Suppose $W([x]_k)[g] \neq W([y]_k)[g]$ for some $0 \leq g < n$. Choose L with $k^L \geq n$ and set $\varphi(Z) = \lfloor nZ/k^L \rfloor$ for $0 \leq Z < k^L$. Then $\varphi(0) = 0$, $\varphi(k^L - 1) = \lfloor n - n/k^L \rfloor = n - 1$ (as $0 < n/k^L \leq 1$), and $\varphi(Z + 1) - \varphi(Z) \leq \lceil n/k^L \rceil = 1$, so φ is onto $\{0, \dots, n - 1\}$. Pick Z with $\varphi(Z) = g$ and let $z \in \Sigma_k^L$ be the zero-padded representation of Z . Applying (2) for both x and y , the $h(w)$ coincide and

$$h(n[xz]_k) - h(n[yz]_k) = W([x]_k)[g] - W([y]_k)[g] \not\equiv 0 \pmod{m},$$

so $x \not\sim y$.

Finally $W([x]_k)$ depends on x only through $[x]_k$, and every $j \geq 0$ equals $[x]_k$ for some x , so the Nerode classes biject with $\{W(j) : j \geq 0\}$. $\square$

Remark 3.2. Lemma 3.1 generalises [6, Thm. 16] (the case $k = m = 2$, n odd), and the proof is theirs. Two points are worth isolating: it holds for *every* n , even or odd, and it needs *no* reachability hypothesis. For digit-additive sequences the Theorem 9(a) construction is thus always optimal; the only remaining question is which windows occur at multiples of n .

4. LEMMA B: FOR ODD n , EVERY FACTOR OCCURS AT A MULTIPLE OF n

Lemma 4.1 (Lemma B). *Let $m \geq 2$, $h = t_{2,m}$, and let $n \geq 1$ be odd. Then every length- n factor of h occurs at some position divisible by n . Consequently, by Lemma 3.1,*

$$\text{sc}(t_{2,m}(ni)) = \rho_{t_{2,m}}(n).$$

Proof. Write $s = s_2$ and fix t with $2^t \geq n$. Every position P factors uniquely as $P = 2^t a + b$ with $0 \leq b < 2^t$. For $0 \leq g < n \leq 2^t$ there are two cases: if $b + g < 2^t$ then $h(P + g) = h(a) + h(b + g)$, while if $b + g \geq 2^t$ then $0 \leq b + g - 2^t < 2^t$ and $P + g = 2^t(a + 1) + (b + g - 2^t)$, so $h(P + g) = h(a + 1) + h(b + g - 2^t)$; both by digit-additivity. Hence the length- n factor at P is a function of the *type*

$$(u, v, b) = (h(a), h(a + 1), b).$$

It therefore suffices, given P of type (u, v, b) , to produce $j \geq 0$ such that nj has the same type.

Step 1: available types at multiples of n . As n is odd it is invertible modulo 2^t ; choose $j_0 \in [0, 2^t)$ with $nj_0 \equiv b \pmod{2^t}$. Then $nj_0 - b$ is a multiple of 2^t exceeding -2^t , so $a_0 := (nj_0 - b)/2^t \in \mathbb{N}$. For $j = j_0 + 2^t j'$, $j' \geq 0$,

$$nj = 2^t(a_0 + nj') + b,$$

so the high parts realisable with this b are exactly the integers $a' \geq a_0$ with $a' \equiv a_0 \pmod{n}$.

Step 2: hitting the pair (u, v) . Choose $T \in \{1, \dots, m\}$ with $T \equiv u - v + 1 \pmod{m}$, choose any $L > T$, and consider

$$a' = 2^L A + (2^T - 1), \quad A \geq 0.$$

Bits $0, \dots, T-1$ of a' are 1 and bit T is 0 (here $L > T$ is used), so $s(a') = s(A) + T$; and $a' + 1 = 2^L A + 2^T$, so $s(a' + 1) = s(A) + 1$, no carry reaching A . Hence if $h(A) \equiv v - 1 \pmod{m}$ then

$$h(a' + 1) = (v - 1) + 1 = v, \quad h(a') = (v - 1) + T \equiv u \pmod{m},$$

as required.

Step 3: the congruence. The constraint $a' \equiv a_0 \pmod{n}$ reads $2^L A \equiv a_0 - 2^T + 1 \pmod{n}$, and 2 is invertible modulo the odd number n , so it is a single congruence $A \equiv A_0 \pmod{n}$. By Gelfond's theorem [4] — for a base $q \geq 2$ and a modulus m with $\gcd(m, q-1) = 1$, the digit sum s_q is equidistributed modulo m along every arithmetic progression; for $q = 2$ the hypothesis reads $\gcd(m, 1) = 1$ and is automatic — there are infinitely many $A \equiv A_0 \pmod{n}$ with $s(A) \equiv v-1 \pmod{m}$. Choosing such an A large enough that $a' \geq a_0$ and setting $j' = (a' - a_0)/n \geq 0$, $j = j_0 + 2^t j'$, the multiple nj has type (u, v, b) and hence carries the same length- n factor as P .

By Lemma 3.1, $\text{sc}(t_{2,m}(ni)) = \#\{W(j) : j \geq 0\}$, and the $W(j)$ are exactly the length- n factors of $t_{2,m}$ at multiples of n ; by the above these are all of them. $\square$

Remark 4.2. Lemma 4.1 is the m -general replacement for [6, Lemma 15], which covers $m = 2$ by citing [2]; it re-proves [6, Thm. 14] without that citation. Oddness of n is used twice (invertibility of $n \pmod{2^t}$, of $2 \pmod{n}$) and cannot be dropped: for even n , $\#\{W(j)\} < \rho_{t_{2,m}}(n)$ strictly. This costs nothing, since $t_{2,m}(2i) = t_{2,m}(i)$ makes $\text{sc}(t_{2,m}(ni))$ depend only on the odd part of n .

5. THEOREM A

The third ingredient is the factor complexity of $t_{2,m}$, determined by Tromp and Shallit [8] and, for all bases, by Starosta [7], whose Table 2 gives, for the base- b word modulo m with q the order of $b-1$ in $\mathbb{Z}/m\mathbb{Z}$ (so $q = m$ for $b = 2$),

$$\begin{aligned} C(n) &= qm(n-1) - m(b^\kappa - b^{\kappa-1}) & \text{for } n = b^\kappa + 1 + \ell, \ 0 \leq \ell < b^\kappa - b^{\kappa-1}, \\ C(n) &= qm(n-1) - m(b^\kappa - b^{\kappa-1} + \ell) & \text{for } n = (2b-1)b^{\kappa-1} + 1 + \ell, \ 0 \leq \ell < b^{\kappa+1} - 2b^\kappa + b^{\kappa-1}. \end{aligned}$$

For $b = 2$ the two regimes merge.

Proposition 5.1. *For $n \geq 3$ let $p \geq 0$ and $0 < r \leq 2^p$ be the unique integers with $n-1 = 2^p + r$. Then $\rho_{t_{2,m}}(n) = m^2(n-1) - m \max(2^{p-1}, r)$, where 2^{p-1} is read as the real number $1/2$ when $p = 0$.*

The reduction is direct: the first regime is $r = \ell < 2^{\kappa-1}$ (with $\ell = 0$ giving $p = \kappa - 1$, $r = 2^{\kappa-1}$), the second $r = 2^{\kappa-1} + \ell \geq 2^{\kappa-1}$; at $m = 2$ this specialises to Brlek's formula [3], whence [6, Cor. 17]. The case $p = 0$ occurs only for $n = 3$, where $\max(1/2, 1) = 1$ and $\rho_{t_{2,m}}(3) = 2m^2 - m$; readers preferring integers may treat $n = 3$ as a base case.

Theorem 5.2 (Theorem A). *Let $m \geq 2$ and let $n \geq 3$ be odd. Write $n-1 = 2^p + r$ with $0 < r \leq 2^p$. Then*

$$\text{sc}(t_{2,m}(ni)) = \rho_{t_{2,m}}(n) = m^2(n-1) - m \max(2^{p-1}, r).$$

Consequently, for every odd $n \geq 3$,

$$\text{sc}(t_{2,m}(ni)) \geq \left(1 - \frac{1}{2m}\right)m^2(n-1) \geq \frac{3}{4}m^2(n-1) \geq \frac{1}{2}nm^2,$$

and along the odd values $n = 3 \cdot 2^{p-1} + 1$ ($p \geq 2$) the state count equals $(1 - \frac{1}{3m})m^2(n-1)$ exactly, so that

$$\limsup_{\substack{n \rightarrow \infty \\ n \text{ odd}}} \frac{\text{sc}(t_{2,m}(ni))}{nm^2} = 1 - \frac{1}{3m} \geq \frac{5}{6}.$$

In particular the family $(t_{2,m})_{m \geq 2}$ answers Problem 1 with $C = 1/2$ — for all odd $n \geq 3$, not merely infinitely many.

Proof. The first equality is Lemma 4.1 (which uses Lemma 3.1), the second is Proposition 5.1.

For the bounds put $N = n - 1 = 2^p + r$. Since $r > 0$, $2^p \leq N$ and so $2^{p-1} \leq N/2$; since $r \leq 2^p$, $N \geq 2r$ and so $r \leq N/2$. Hence $\max(2^{p-1}, r) \leq N/2$, with equality iff $r = 2^p$, and $\text{sc} \geq m^2 N - mN/2 = (1 - \frac{1}{2m})m^2 N \geq \frac{3}{4}m^2 N$ as $m \geq 2$; finally $N \geq \frac{2}{3}n$ for $n \geq 3$, giving $\text{sc} \geq \frac{1}{2}nm^2$.

For the limsup, minimise $f(p, r) = \max(2^{p-1}, r)/(2^p + r)$ over $0 < r \leq 2^p$: on $r \leq 2^{p-1}$ it is $2^{p-1}/(2^p + r)$, decreasing in r , and on $r \geq 2^{p-1}$ it is $r/(2^p + r)$, increasing in r , so the minimum is $f = 1/3$ at $r = 2^{p-1}$. Thus $\max(2^{p-1}, r) \geq (n-1)/3$ always, with equality exactly when $n = 3 \cdot 2^{p-1} + 1$ (odd precisely for $p \geq 2$). Hence $\text{sc}(t_{2,m}(ni)) \leq (1 - \frac{1}{3m})m^2(n-1) < (1 - \frac{1}{3m})nm^2$ for every odd $n \geq 3$, with equality in the first inequality along $n = 3 \cdot 2^{p-1} + 1$, where $(n-1)/n \rightarrow 1$; and $1 - 1/(3m) \geq 5/6$ for $m \geq 2$. $\square$

Remark 5.3. Against the bound actually proved in [6], namely $2nm^2$ for $k = 2$, the family realises the fraction $(1 - \frac{1}{3m})/2 \in [\frac{5}{12}, \frac{1}{2}]$: Corollary 10(a) is tight in the shape nm^2 , loose only in the factor k (cf. Theorem 7.1).

6. VERIFICATION

Theorem 5.2 is a proof, conditional only on the two literature inputs (Proposition 5.1 and Gelfond's theorem). It was nevertheless checked numerically by implementations sharing no code. Two computed sc directly: a *theory-free* msd Myhill–Nerode search, in which the signature of X is the raw table $(u(k^L X + j))_{L \leq L_{\max}, j < k^L}$ built from the oracle $u(i) = s_2(ni) \bmod m$ alone, with transitions $X \mapsto kX + d$ (distinct signatures are provably distinct states, so it cannot over-count); and a breadth-first construction of the Theorem 9(a) window automaton followed by Moore minimisation. Separately, $\rho_{t_{2,m}}(n)$ was computed by suffix doubling on a 2^{22} -symbol prefix, and $\#\{W(j)\}$ straight from population counts, with no automaton involved. A further independent code path is Peanut, which compiles the predicate $\mathbf{T}[n*i]=a$ into a minimal DFA.

Check	Scope	Result
Nerode search vs. window construction vs. Thm. 5.2	$m = 2 \dots 7$, odd $n = 3 \dots 101$ (300 cells)	no mismatch
Thm. 5.2 vs. window construction, adversarial parameters	$m \in \{8, 10, 12, 16, 20\} \times 7$ values of n ; 12 further cells up to $m = 32$	no mismatch
$\rho_{t_{2,m}}$ by suffix doubling vs. Prop. 5.1	$m = 2 \dots 7$, every n from 3 to 130	no mismatch, no non-convergence
$\#\{W(j) : j < J\}$ from population counts vs. $\rho_{t_{2,m}}(n)$ (Lem. 4.1)	slowest-converging cells, $J < 2 \cdot 10^7$	exact, e.g. $(m, n) = (7, 101) \mapsto 4648$
Peanut, $\mathbf{dfa} \ \mathbf{T}[n*i]=a$	14 cells, $m \leq 7$, $n \leq 101$	agrees with both constructions
subsequence $n = 3 \cdot 2^{p-1} + 1$, $p = 2 \dots 6$	$m \in \{2, 3, 5, 7\}$ (20 cells)	$\text{sc} = (1 - \frac{1}{3m})m^2(n-1)$ exactly
extremes of $\text{sc}/(nm^2)$, $\text{sc}/(m^2(n-1))$	the 300 cells above	minima $1/2$ (at $m = 2, n = 3$) and $3/4$

Two caveats about status. First, reading the state count of the per-letter DFA $\mathbf{dfa} \ \mathbf{T}[n*i]=a$ as sc presumes each level set $\{i : t_{2,m}(ni) = a\}$ needs the full sc states. This held in every cell run — all m letters gave the same count, equal to sc — but it is an empirical bridge, not a proof, so the engine figures corroborate the two constructions rather than replace them. Second, and more importantly:

Observation 6.1 (base 3; machine-verified only). Let $g_m(i) = (\#\{\text{digits equal to 2 in } (i)_3\}) \bmod m$, a 3-digit-additive sequence with weights $(0, 0, 1)$, generated by a minimal m -state zero-consistent base-3 DFAO. Then $\text{sc}(g_m(ni)) = \rho_{g_m}(n) = m^2(n-1)$ exactly whenever $3 \nmid n$, and $\text{sc}(g_m(ni)) = \text{sc}(g_m(n'i))$ for $n = 3^v n'$ with $3 \nmid n'$.

Observation 6.1 was reproduced independently for $m = 2 \dots 6$ and $n \in \{2, 4, 5, 7, 8, 11, 13, 16, 22, 31, 40, 44\}$, and is *not* proved here. Lemma 3.1 applies verbatim, so $\text{sc}(g_m(ni)) = \#\{W(j)\}$; the missing ingredients are an analogue of Lemma 4.1 — Gelfond's hypothesis $\gcd(m, q-1) = 1$ fails for $q = 3$ and even m , though the sequence here is a weighted digit sum rather than s_3 — and a proof that $\rho_{g_m}(n) = m^2(n-1)$. This family saturates nm^2 , reaching $\text{sc}/(nm^2) = 1 - 1/n$, but only one third of the knm^2 of Corollary 10(a). It is recorded as a conjecture, not a result.

7. REMARKS

A sharpening of Corollary 10(a). The window parametrisation of Lemma 4.1 is not special to $t_{2,m}$, and gives the following, in which the factor k of Corollary 10(a) becomes $1 + (k-1)/m$.

Theorem 7.1 (Theorem C). *Let h be generated by an m -state zero-consistent msd-first DFAO with interior sequence h' and intrinsic DFAO $\tilde{M}$, let $n \geq 1$, and let t be minimal with $k^t \geq n$. Then*

$$\rho_{h'}(n) \leq \rho_{\tilde{M}}(2)(n-1) + m(k^t - n + 1) \leq m^2(n-1) + m((k-1)n + 1),$$

and hence $\text{sc}(h(ni + c)) \leq m^2(n-1) + m((k-1)n + 1)$ for all $0 \leq c < n$.

Proof. Write $P = k^t a + b$, $0 \leq b < k^t$. Every position in $[P, P + n)$ is $k^t(a + e) + r$ with $e \in \{0, 1\}$, $0 \leq r < k^t$, and $h'(k^t A + r) = \delta(h'(A), r_1 \cdots r_t)$ where $r_1 \cdots r_t$ is r zero-padded to length t , since msd reading concatenates digit blocks — exactly where zero-consistency is needed. So the factor at P is determined by $(h'(a), h'(a + 1), b)$, and by $(h'(a), b)$ alone when $b + n \leq k^t$. There are $k^t - n + 1$ values of b of the latter kind, each contributing at most m factors, and $n - 1$ of the former, each contributing at most the number $\rho_{\widehat{M}}(2)$ of realised pairs $(h'(a), h'(a + 1))$. The second inequality uses $\rho_{\widehat{M}}(2) \leq m^2$ and $k^t \leq k(n - 1) \leq kn - 1$, the latter by minimality of t . $\square$

Zero-consistency is a genuine hypothesis, not a normalisation: without it $h'(0) = q_0$ fails, the recursion $h'(ki + b) = \delta(h'(i), b)$ breaks at $i = 0$, and the window construction of [6, Thm. 9] — hence Theorem 7.1 and every window count in this note — is invalid. Both families here are zero-consistent, as is every digit-additive sequence. Theorem 7.1 improves knm^2 when $m \gg k$, but it is a one-shot form of the standard blocking argument behind [1, Thm. 10.3.1] and should be regarded as folklore-grade.

On significance. This note assembles three known pieces rather than proving a new hard theorem. The proof of [6, Thm. 16] generalises essentially verbatim to Lemma 3.1; their Lemma 15 needs a replacement, supplied by the short construction of Lemma 4.1 with Gelfond’s 1968 theorem; and the growth rate $m^2 n$ is already in [8, 7]. What was absent from the source is precisely the join of the three. The answer appears correct and complete, but should be read as a routine composition, and may well be known to specialists in a form we have not located.

The constant is not improvable uniformly. Within this family $C = 1/2$ is exactly attained, at $(m, n) = (2, 3)$, where $\text{sc} = 6 = \frac{1}{2} \cdot 3 \cdot 2^2$. So $1/2$ is the largest constant valid for *all* odd $n \geq 3$ and all $m \geq 2$ here, and the stronger $1 - 1/(3m)$ of Theorem 5.2 is a statement about a subsequence of n only. Since Problem 1 asks only for infinitely many n , either constant answers it.

What is not settled. Three questions remain open. (i) Observation 6.1: the identity $\rho_{g_m}(n) = m^2(n - 1)$ and an analogue of Lemma 4.1 in base 3. (ii) The exact maximum of $\text{sc}(h(ni))$ over all m -state k -DFAOs. The bound $m^2(n - 1)$ holds exhaustively for all zero-consistent 2- and 3-state DFAOs at $k \in \{2, 3\}$, $n \in \{7, 11, 16, 31\}$, but fails for $k = 4$: weights $(0, 0, 1, 0)$ modulo 3 give $\text{sc}(h(31i)) = 312 > 270 = m^2(n - 1)$, still below Theorem 7.1’s bound of 372. The truth thus lies between $m^2(n - 1)$ and $m^2(n - 1) + m((k - 1)n + 1)$, and the factor k in knm^2 is a real but bounded effect that vanishes as m grows. (iii) The case $c > 0$, i.e. $(h(ni + c))_{i \geq 0}$ and [6, Cor. 10(b)]: only $c = 0$ is treated here, that being what Problem 1 asks.

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